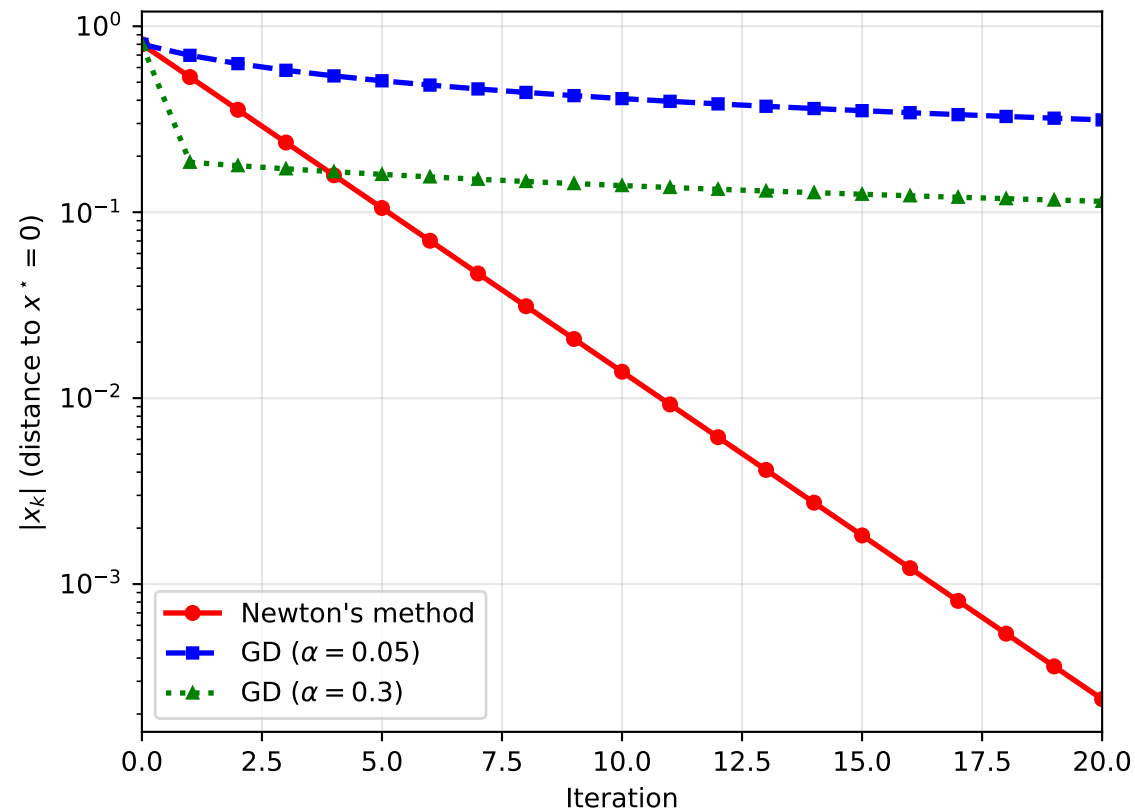


Newton vs GD on $g(x) = x^4$
(superlinear convergence near $x^* = 0$)



Quadratic Loss: Newton One-Step Convergence
 f^* reached after $k = 1$ (quadratic loss)

